



TELECOM CUSTOMER CHURN PREDICTION & TIERED RETENTION STRATEGY

Company A — Wireless Telecommunications Provider

Youssef Sayed | Date: June 18, 2026

Market Analysis



Global Market Size

\$1.8 Trillion global telecom market value in 2024.



Mobile Penetration

Exceeds 100% in most developed markets worldwide.



Acquisition Cost

5–25x higher than customer retention costs (HBR).



Annual Churn Rate

15–25% average annual churn across telecom operators.



A 5% increase in retention can boost profits by 25–95% (Bain & Company)

THE PROBLEM: CUSTOMER CHURN

- **Company A has 100,000 subscribers**
- **49.6% of customers have churned — nearly half the base**
- **This results in an estimated \$34.4 million annual revenue loss**
- **We need to answer: Who will churn next, and how do we stop them?**

Our Goal: To proactively identify which customers are likely to leave before they actually do, enabling us to intervene effectively.



Our Data Foundation

Data Insights

What the data covers: • Demographics & Location • Equipment & Phone Age • Usage & Monthly Billing • Customer Care & Drops



Our Dataset

100K+

Two datasets merged (Client.csv & Record.csv) creating a robust foundation of 100,000 rows and 100 features.

Data Quality: Handled 43 columns with missing values (dropped features with >40% missing). Target Variable: churn (0 = Retained, 1 = Churned).

EDA: Churn Distribution

Revenue Impact

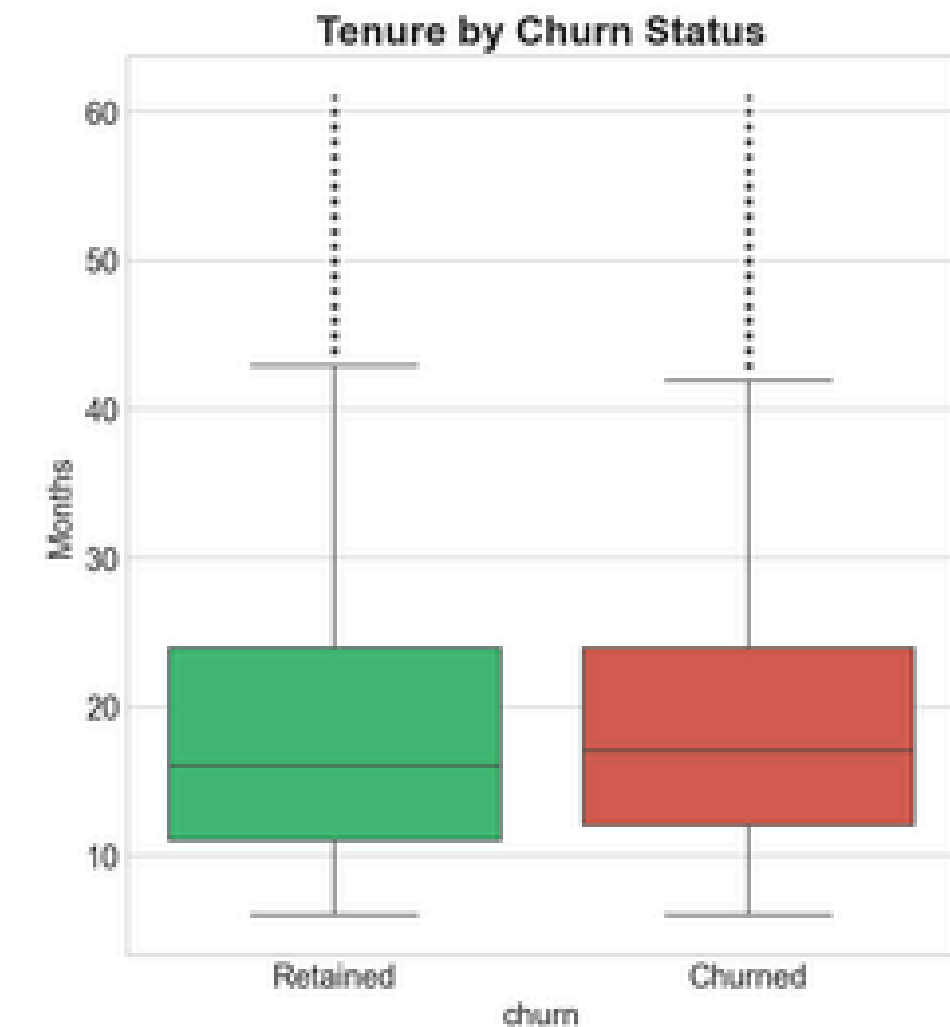
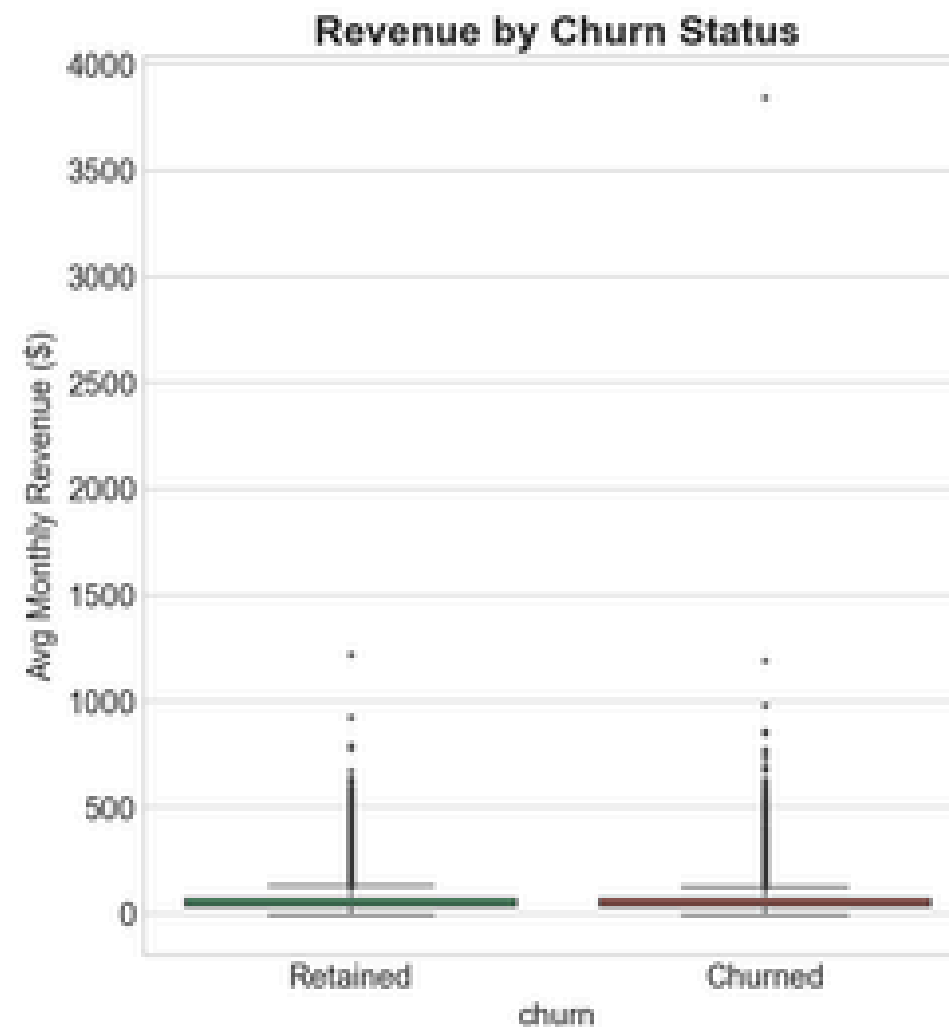
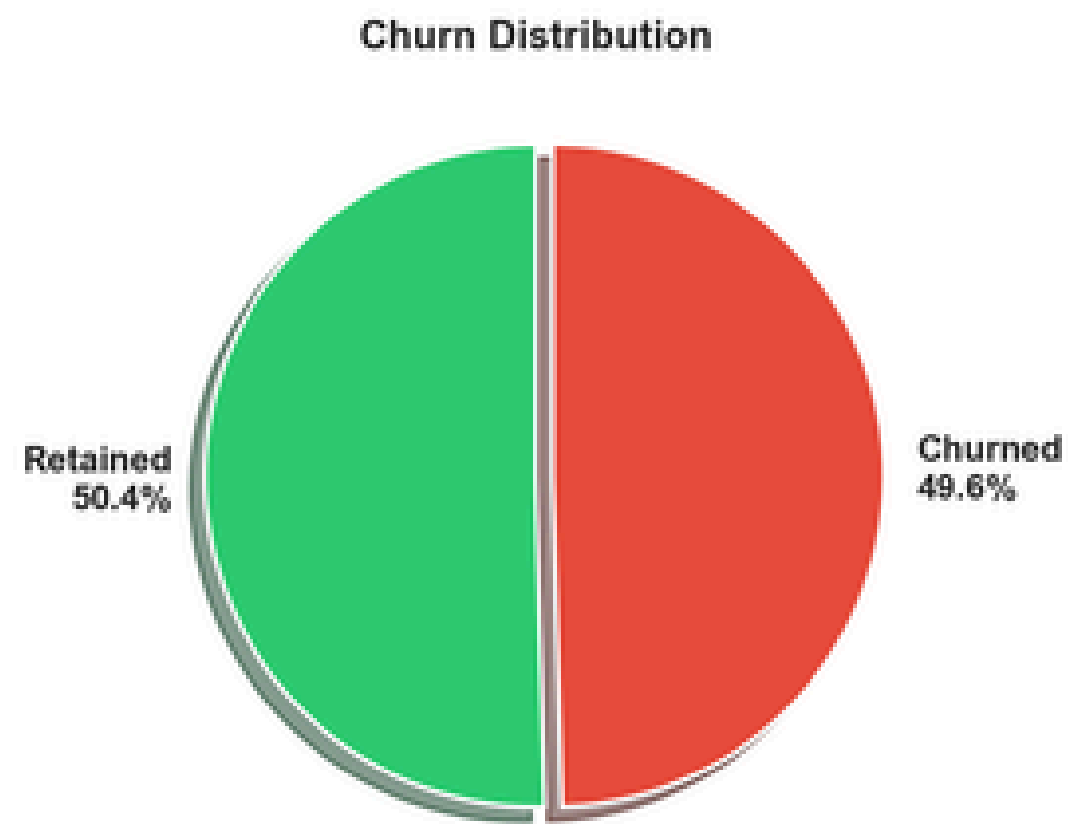
5%~
Revenue Gap

49.6%
Churned

50.4%
Retained

- The churn rate is nearly balanced: 50.4% retained vs. 49.6% churned.
- Churned customers generate slightly lower average monthly revenue compared to retained ones.
- Customer tenure (months with the company) shows no significant difference between the two groups.

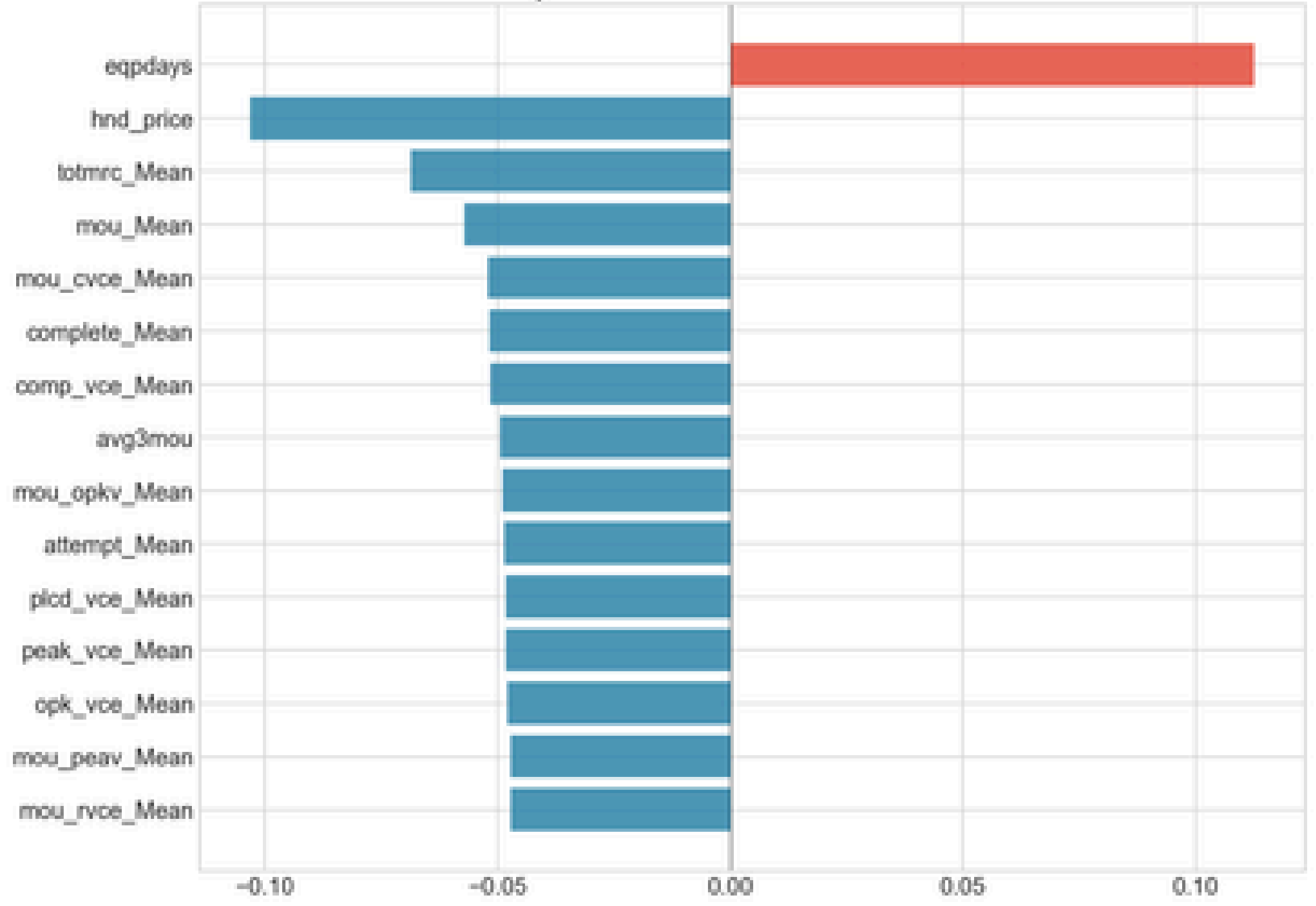
EDA Analysis



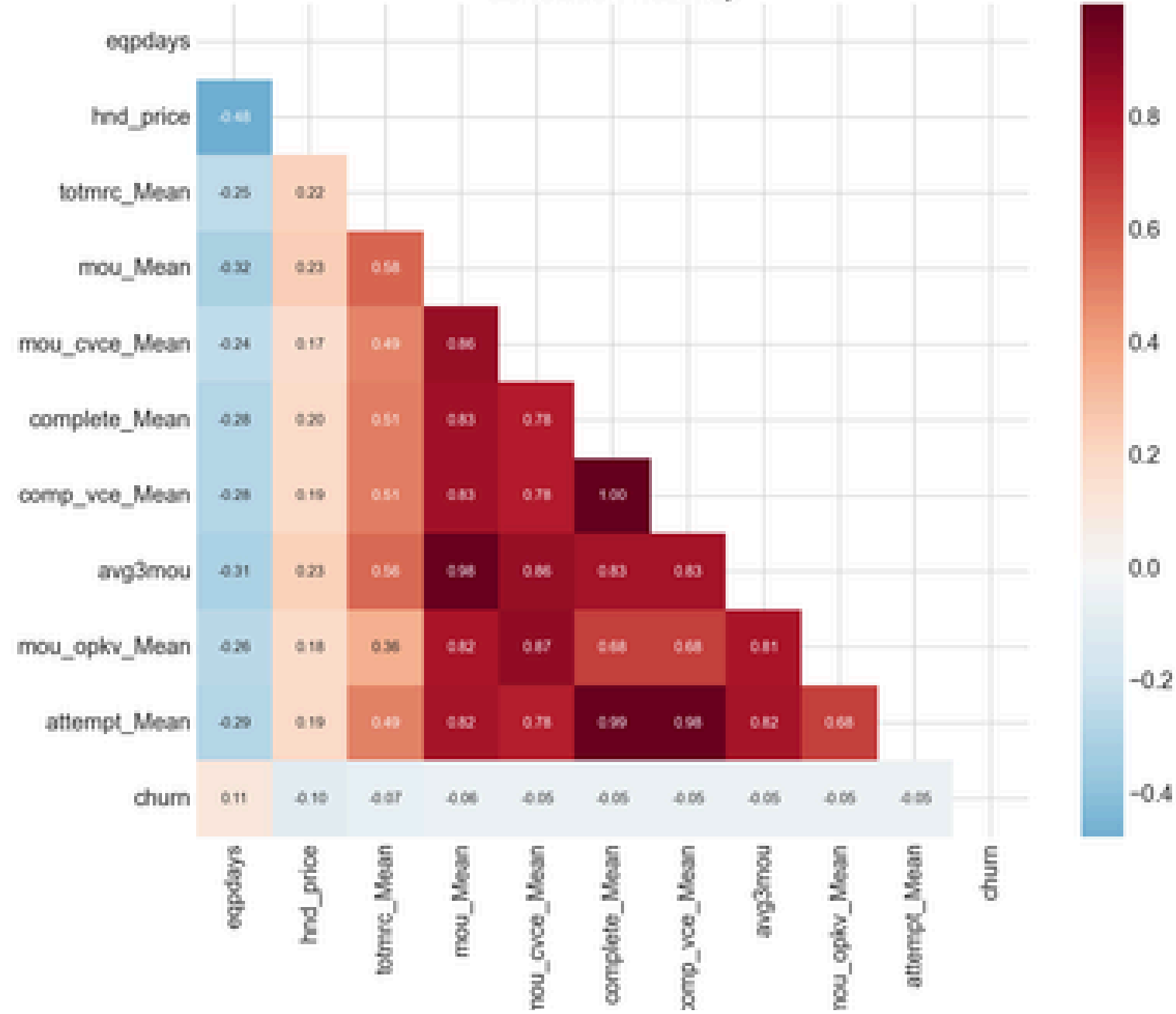
EDA: Feature Correlation with Churn

Correlation analysis reveals the key factors influencing a customer's decision to leave. These findings help prioritize features for our predictive model and guide our feature engineering process.

Top 15 Features Correlated with Churn



Correlation Heatmap



Top Predictors: change_mou is the (change in usage), strongest indicator of churn while eqpdays (equipment age) also shows a significant correlation.

Financial Impact: Revenue and billing metrics show moderate correlation with churn. We will use these insights to engineer new value-based features.

Feature Engineering — 24 New Predictive Signals

Usage Trends ✓

Value & Revenue ✓

Customer Engagement ✓

Network Quality ✓

FEATURE	DESCRIPTION	CATEGORY
revenue_per_minute	Revenue / Minutes of Use	Value
drop_rate	Dropped calls / Total attempts	Quality
usage_change_pct	% change in usage over time	Trend
equip_age_months	Equipment age in months	Equipment
custcare_per_month	Care calls per month	Satisfaction
avg3_vs_avg6_mou	Recent vs long-term usage	Trend
unanswered_rate	Unanswered / Placed calls	Engagement
rev_change_ratio	Revenue change ratio	Trend



Why Feature Engineering? We didn't just rely on raw data. By combining variables, we created 24 new predictive signals across 5 key categories

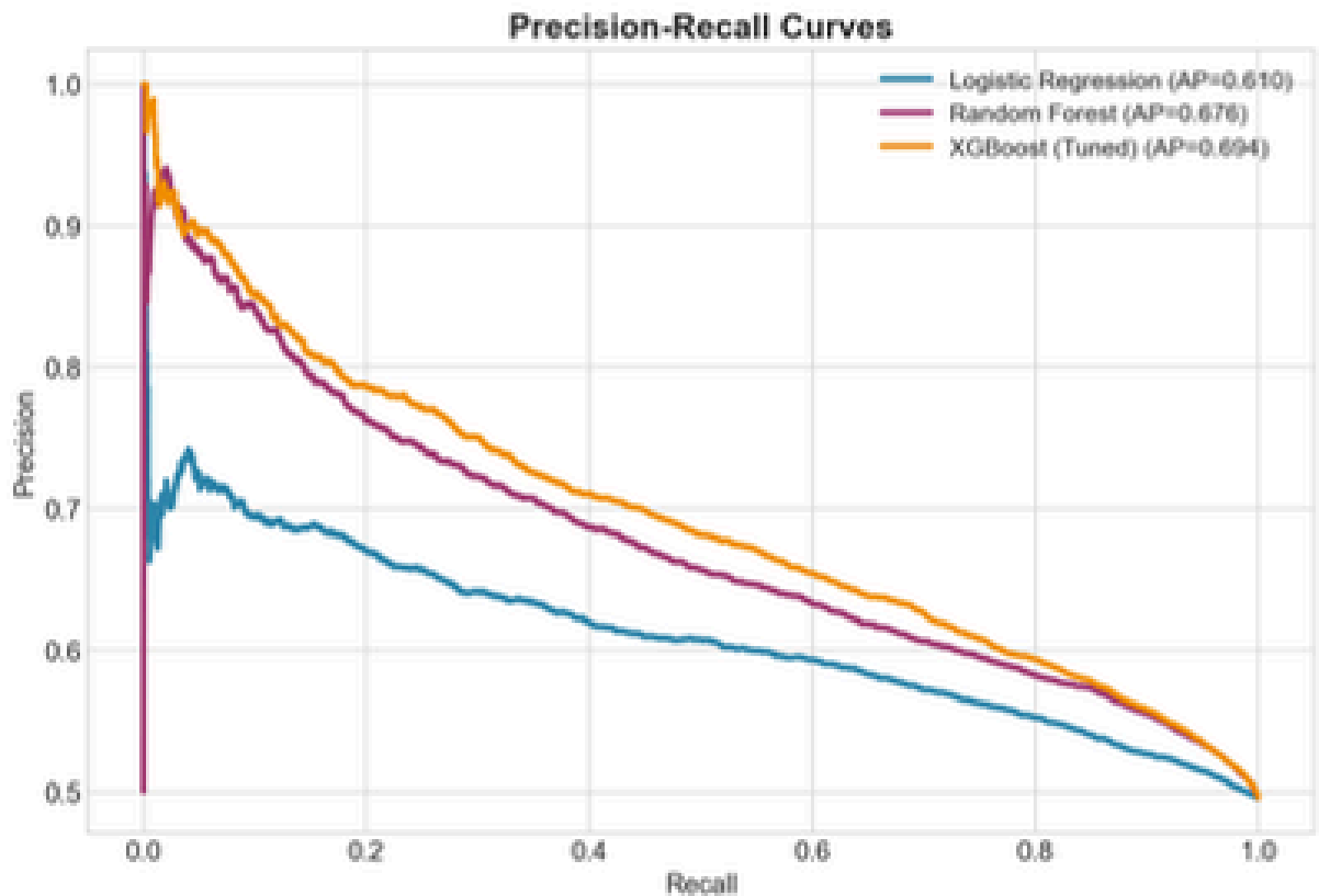
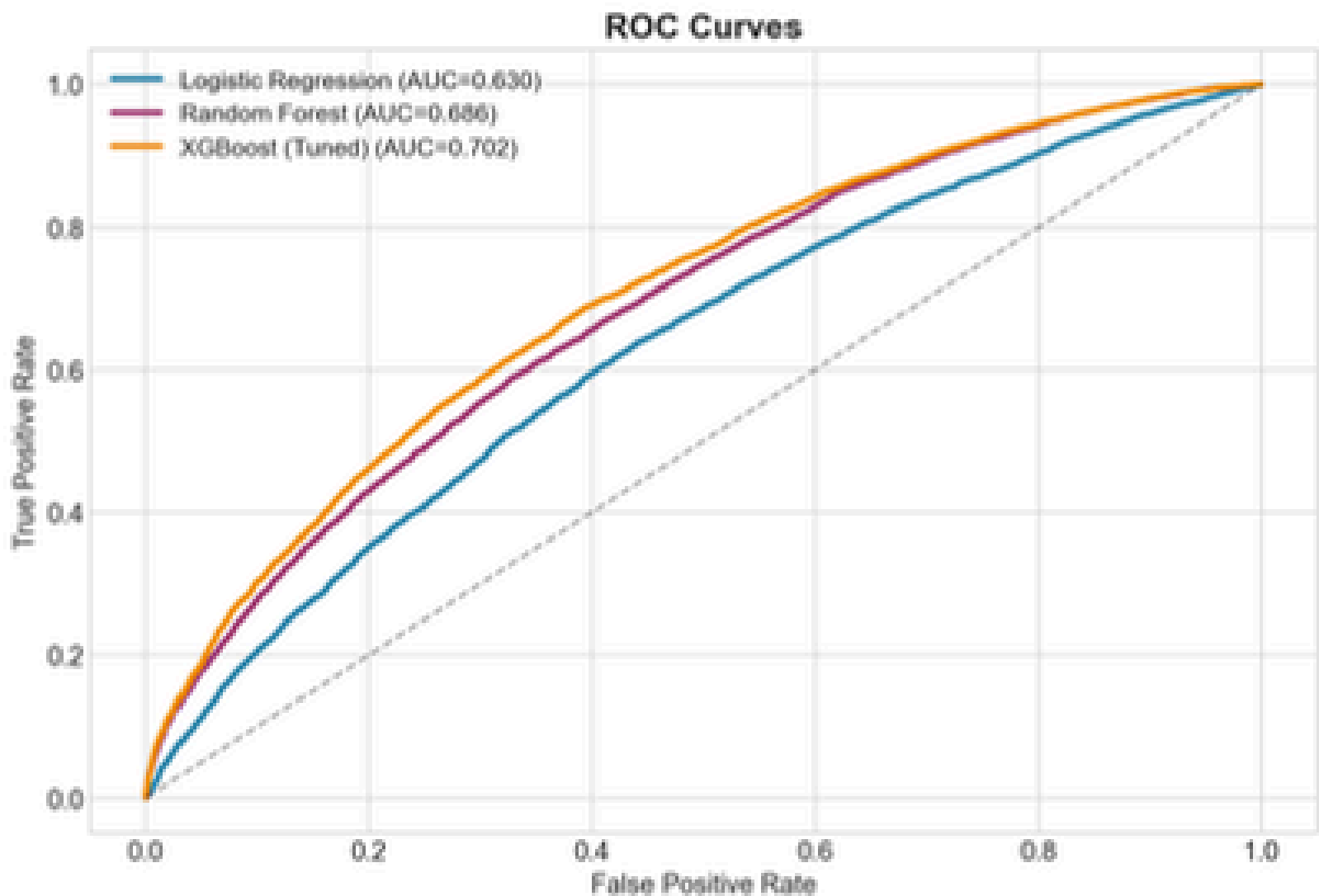
These engineered features (like `usage_change_pct` and `revenue_per_minute`) capture behavioral nuances that significantly improved our .model's ability to predict churn

Model Building — 3 Algorithms Compared

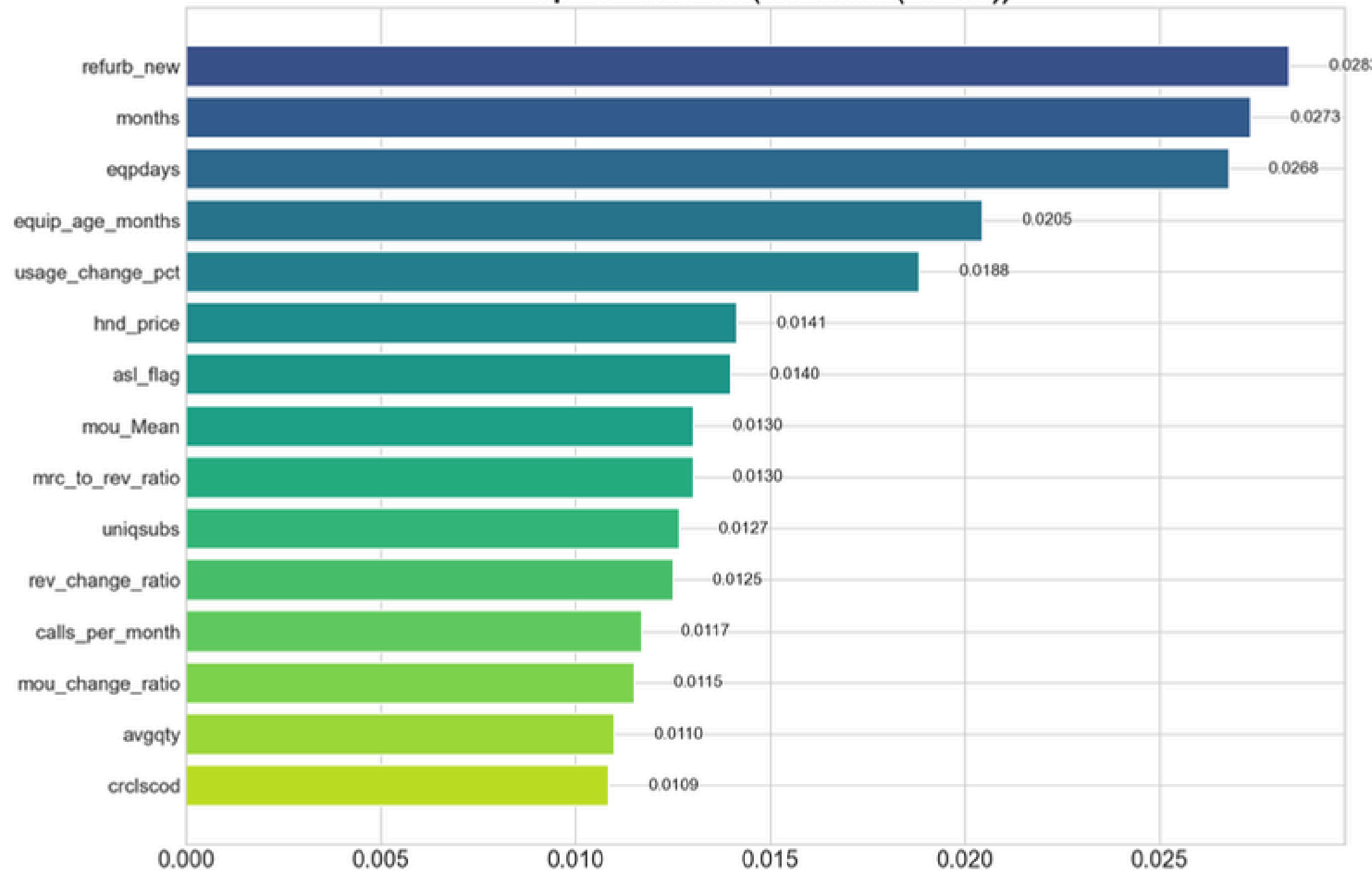
Best Model

Winning Model: XGBoost (Tuned) F1 Score = 0.6416

MODEL	ACCURACY	F1 SCORE	PRECISION	RECALL	CV F1
Logistic Regression	0.5964	0.5870	0.5957	0.5785	0.5863
Random Forest	0.6250	0.6247	0.6240	0.6255	0.6280
XGBoost (Tuned)	0.6440	0.6416	0.6438	0.6394	0.6411



Top 15 Features (XGBoost (Tuned))



Top Drivers for Predictions (Feature Importance)



Key Insights from the Model:

- Equipment Matters:** refurb_new and eqpdays (equipment age) are among the most critical predictors. Customers with older or refurbished phones are highly sensitive to churn.
- Usage Changes:** usage_change_pct strongly signals customer dissatisfaction or disengagement before they leave.
- Financial Sensitivity:** Billing metrics like hnd_price and revenue ratios heavily influence the decision to stay or leave.

Customer Segmentation — Value × Risk Matrix



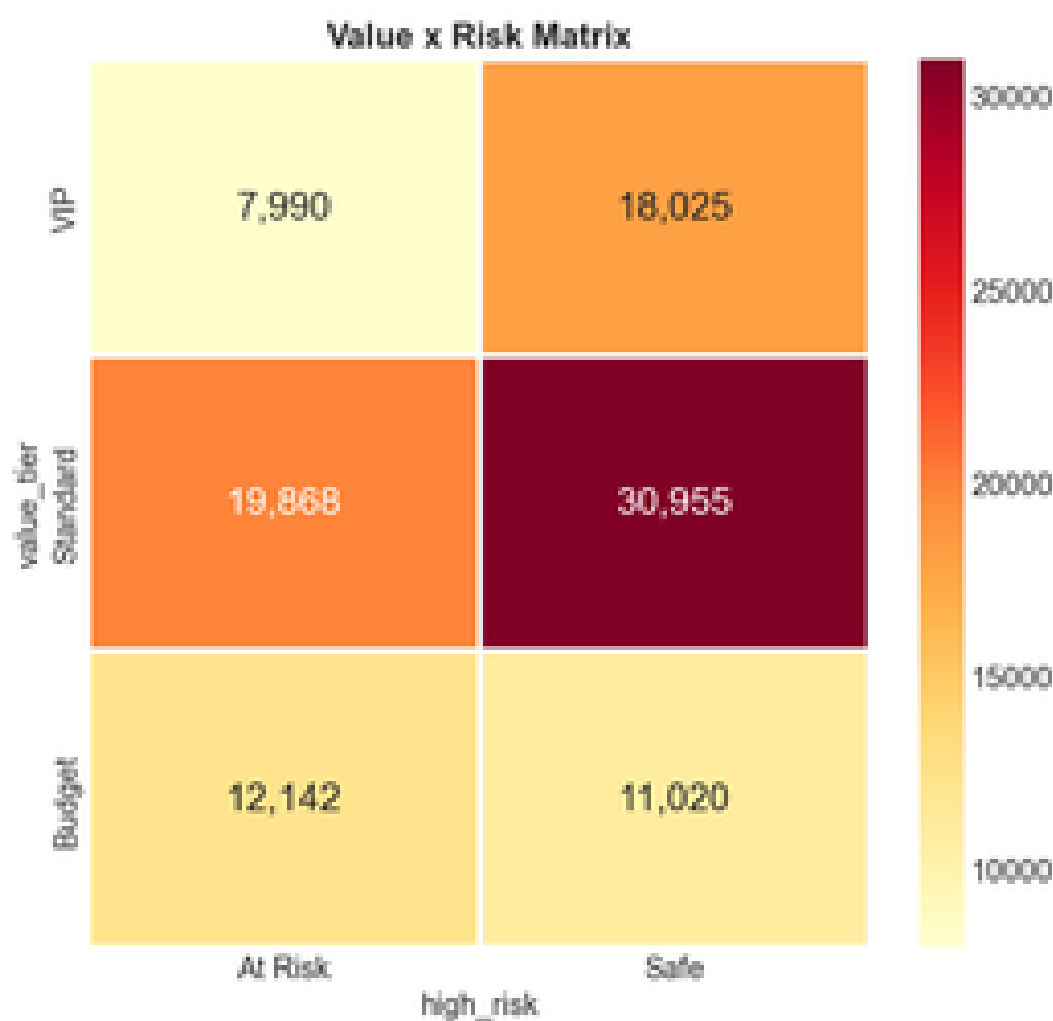
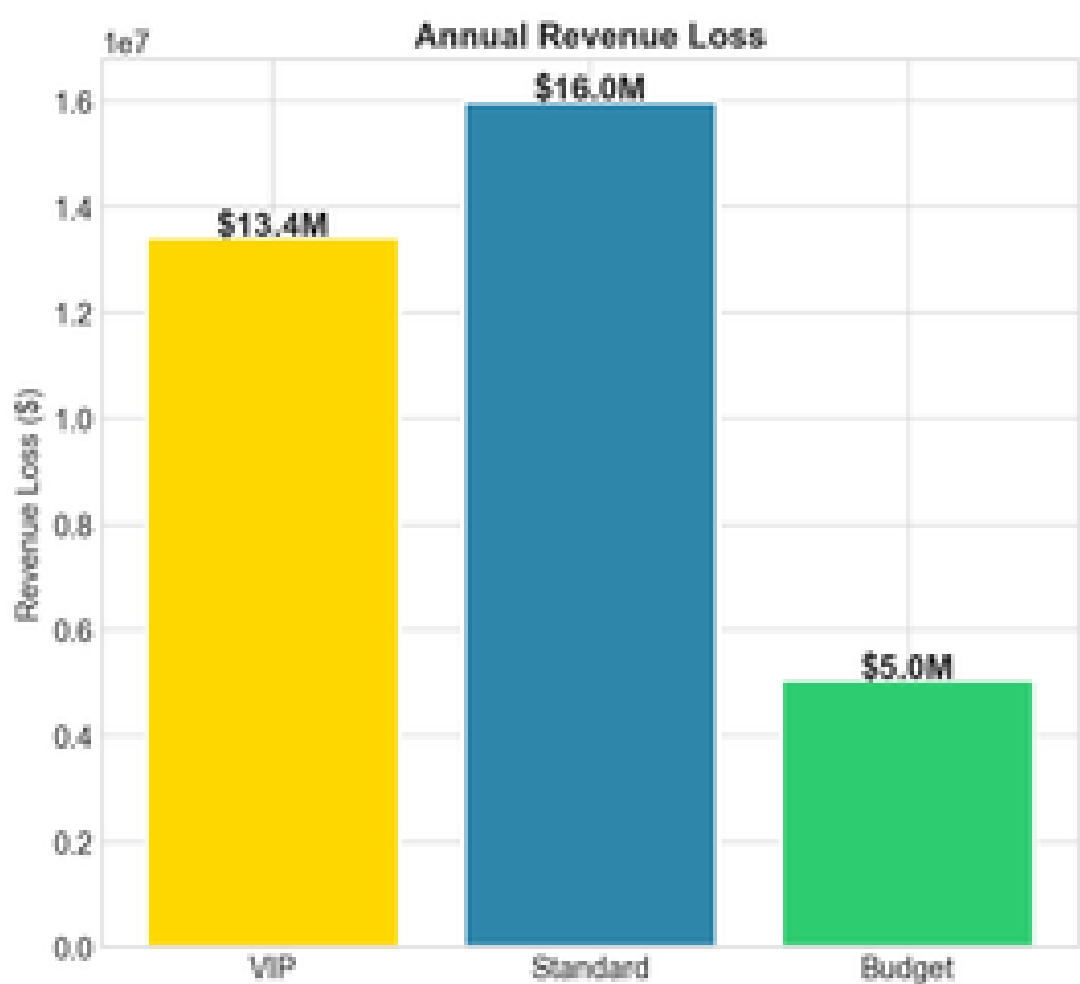
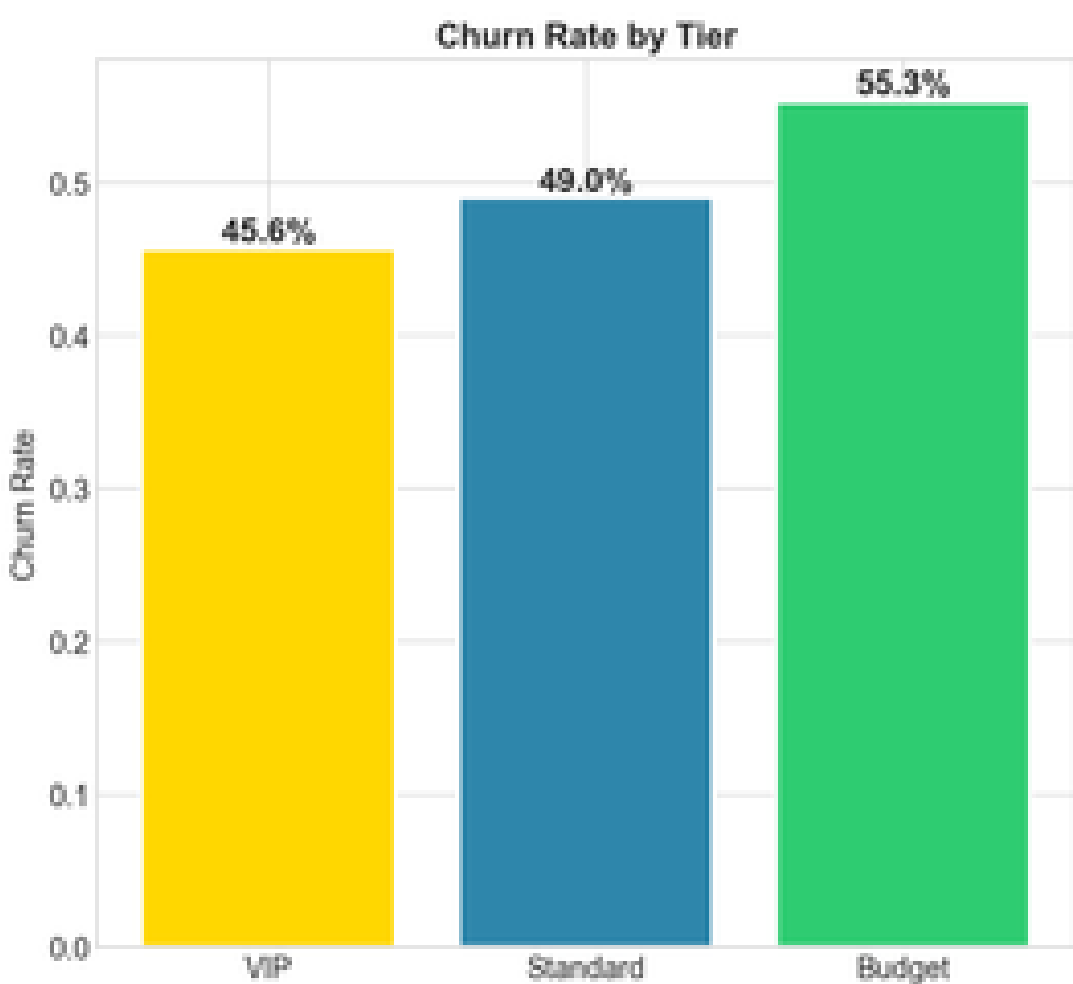
Segment Definitions & Insights:

VIP Tier ($\geq \$60/\text{mo}$): High Value, Moderate Risk (45.6% Churn)
Largest driver of revenue loss if ignored.

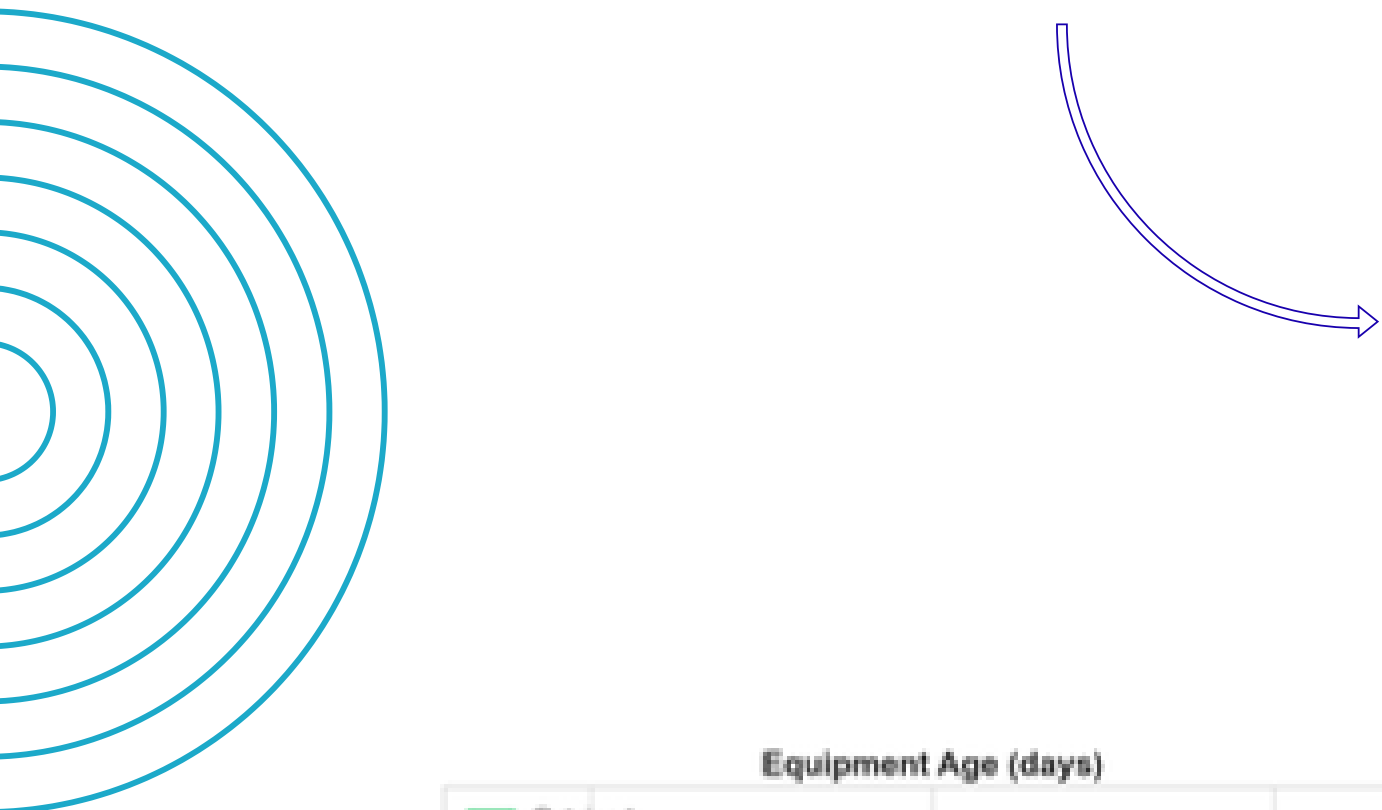
Standard Tier (30-30-60/mo): The core customer base (50K+ users). 49% Churn rate.

Budget Tier ($< \$30/\text{mo}$): Highest Churn Rate (55.3%).

Lowest revenue per user, requiring low-cost retention methods.



Tiered Retention Strategy — Personalized Interventions





VIP Tier

\$150 / customer

- ✓ Dedicated account manager
- ✓ Free router/device upgrade
- ✓ Speed boost + streaming bundle
- ✓ Priority support line

Target Save Rate
30%



Standard Tier

\$50 / customer

- ✓ Free plan upgrade for 3 months
- ✓ Loyalty points program
- ✓ Annual contract with 10% discount
- ✓ Trade-in credit for device

Target Save Rate
25%



Budget Tier

\$10 / customer

- ✓ 15% bill discount (SMS offer)
- ✓ Free bonus data package
- ✓ Family plan: 2nd line at half price
- ✓ "Pay on time" rewards

Target Save Rate
40%



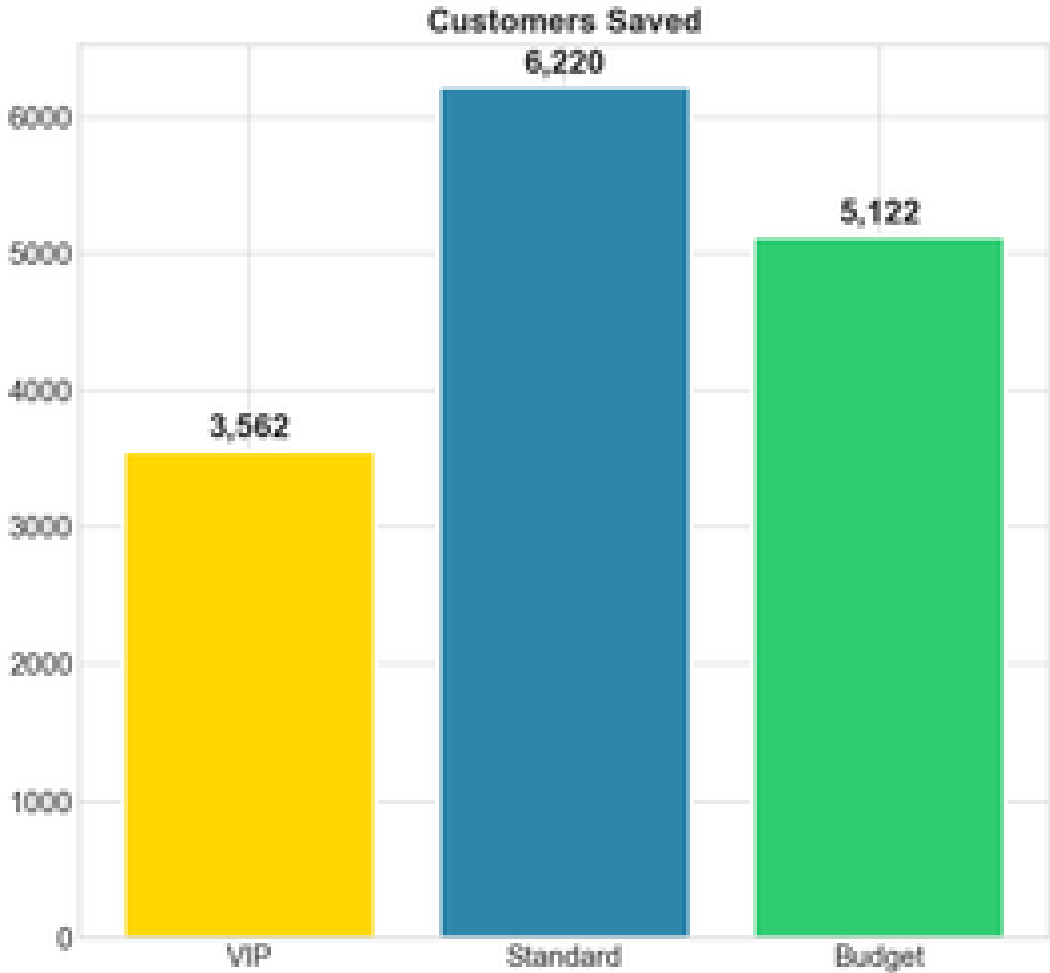
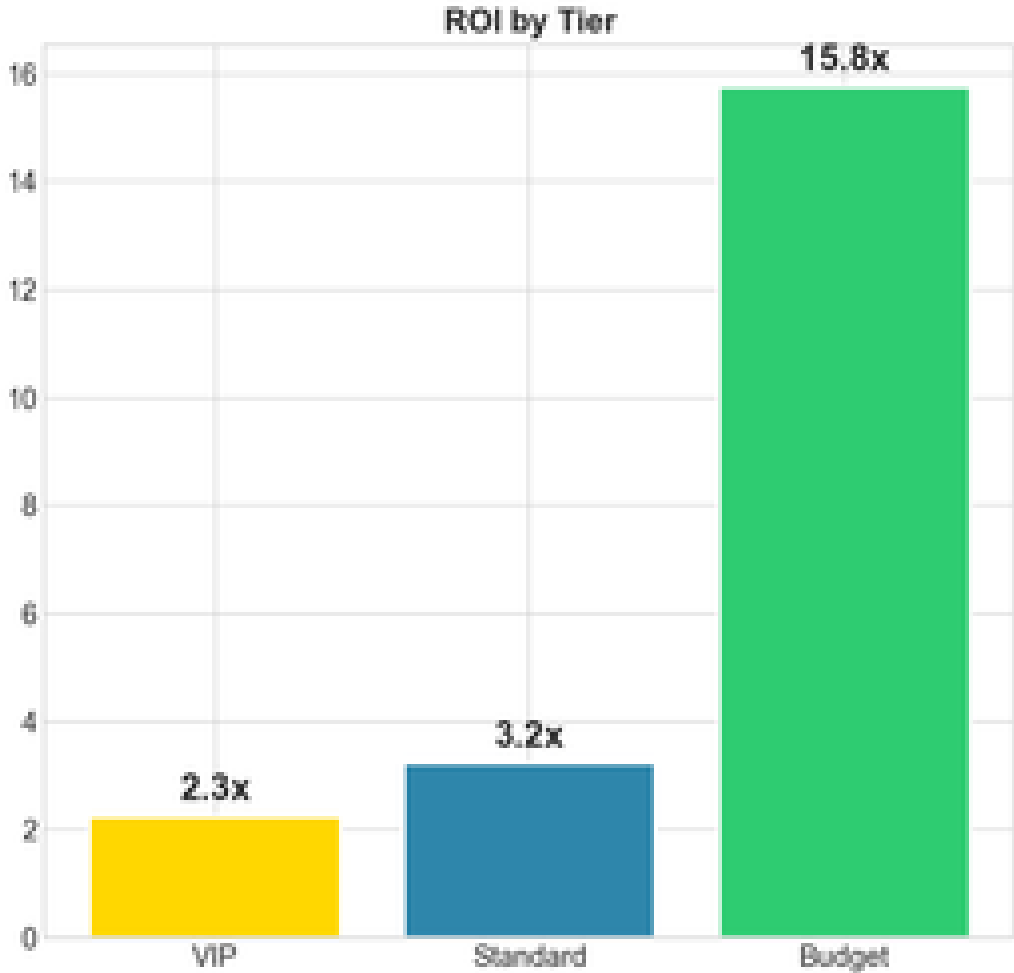
Financial Impact — ROI Analysis

Net Profit: \$6.9 Million / Year

By investing

3.2M**intargetedinterventions,wecanprotect**3.2M**intargetedinterventions,wecanprotect**10.1M in annual revenue. Overall ROI = 3.2x

TIER	INVESTMENT	CUSTOMERS SAVED	REVENUE PROTECTED	
VIP	\$1,781,100	3,562	\$4,023,847/yr	2.3x
Standard	\$1,244,100	6,220	\$4,037,015/yr	3.2x
Budget	\$128,060	5,122	\$2,018,093/yr	15.8x
TOTAL	\$3,153,260	14,904	\$10,078,955/yr	3.2x



Customer Acquisition Strategy

Fueling Growth Beyond Retention



Why Focus on Acquisition? While our ML model successfully protects existing revenue by predicting churn, sustainable business growth requires a constant influx of new customers.

These 5 targeted strategies are designed to capture new market segments, lower customer acquisition costs (CAC), and build long-term loyalty from day one.

STRATEGY	TARGET	DETAILS
 "Try Free" Plan	Competitor users	1 month free trial → convert to paid
 Referral Program	Happy customers	"Bring a friend → both get 25% off"
 Entry-Level Plans	Price-sensitive	\$15/month basic plan → upsell later
 Family Bundles	Households	+\$10/additional line → hard to churn
 Student Plans	Age 18-25	Discounted plan → lifetime customer

Conclusion & Strategic Roadmap

Key Results



F1 = 0.64

Model accurately identifies at-risk customers



\$10.1M Protected Annually

Achieved with just \$3.2M strategic investment



3.2x Overall ROI

Every \$1 invested returns \$3.20 in saved revenue

Next Steps

- 1 Deploy ML model into the monthly scoring pipeline
- 2 Integrate risk scores with CRM for automated outreach
- 3 A/B test tier-specific offers to optimize save rates
- 4 Track ROI quarterly and continuously refine the strategy
- 5 Expand predictive modeling to acquisition campaigns

Every customer matters, but not every customer needs the same offer. Predictive AI makes retention profitable.

References

- 🔍 1. Telecom Churn Dataset — Provided by course instructor.
- 🔍 2. XGBoost — [xgboost.readthedocs.io](https://xgboost.ai)
- 🔍 AI Tool: Antigravity (Gemini Agent) — Code optimization & readability.
- 🔍 <https://gemini.google.com/share/72bc3b704b81> Gemini AI — Assisted with presentation content and slide design.